

Traction Control and Torque Vectoring

IITB Racing — Vehicle Dynamics

One of the biggest advantages of having an EV instead of an IC car is that we can implement Torque Vectoring and Traction Control, which can help us improve our car in the following ways:

- TV allows us to exploit the available extra grip in the outer wheel by transmitting more torque to the outer wheel and less torque to the inner wheel, so that both tires can work well within their limits and still generate the same amount of driving moment.
- A second effect of TV is that it improves the cornering dynamic response of the vehicle. The yaw moment generated by torque vectoring also acts as a way to control the tire sideslip angle of the rear tires — with TV, the rear tires generate higher forces more quickly than the rear tires of a passive vehicle.
- TC can be used beyond just managing wheel slip, by independently modulating torque to each motor. It generates a net yaw moment that actively corrects the car's yaw response, driving the vehicle toward a desired yaw rate — effectively achieving neutral steer — and can be tuned in real time based on driver feedback to suit their preference and driving style.

The parameters and aspects we can control or tune with TC and TV depend heavily on the sensors we have and the data we can collect while running the car.

Available Sensors

- Both rear left and right motor torque and RPM, provided by the 2 inverters
- Steering angle, measured by a rotational potentiometer
- Vehicle acceleration and yaw rate, based on data from 1 inertial measurement unit (IMU)

Based on the data we can record while running the car, and the controllable parameters available, we decided to control the yaw rate of the car. Controlling and correcting the yaw rate helps us get closer to a neutral-steer condition, and if required, this can be tuned according to the driver's feedback.

Desired Yaw Rate

To arrive at the desired value of the yaw rate — in simple terms, “what the yaw rate should be” — based on the driver's steering inputs and the current vehicle velocity, we take the following approach:

- It is assumed that the vehicle is in a steady-state condition.
- A neutral-steer vehicle has the smallest possible turning radius for a given velocity, which corresponds to optimal performance. Therefore, a neutral-steer vehicle is chosen as the reference.
- The ideal, or desired, yaw rate can be defined by the velocity and the radius of the curve:

$$\dot{\psi}_{desired} = \frac{v_{CG}}{R} \quad (3.5)$$

Given the velocity and steering angle of the car and with the known steer gradient and wheelbase it is possible to know the turning radius.

$$\frac{1}{R} = \frac{\delta}{(l_r + l_f) + K_u v_{CG}^2} \quad (3.6)$$

Combining both expressions 3.5 and 3.6, the under-steer gradient and the circular road radius gives the desired yaw reference.

Eq. 3.5, 3.6 — desired yaw rate and turning radius from steer gradient and wheelbase

Combining expressions 3.5 and 3.6, the understeer gradient and the circular road radius give the desired yaw reference:

$$\dot{\psi}_{desired} = \frac{v_{CG}}{(l_r + l_f) + K_u v_{GC}^2} \delta$$

Combined desired yaw-rate reference

The yaw rate of the car is constantly measured by the IMU, the steering angle by the steering potentiometer, and the instantaneous velocity is obtained as follows: the inverters at the motor controllers on both rear tires give the angular velocity of the rear left and right motors. We take the average of the two, divide by the gear ratio (4.55), and multiply by the tire radius (0.2286 m) to obtain the instantaneous velocity of the car.

This approach is not the most accurate way to measure instantaneous velocity, but given the sensors currently available to us, it was the most sought-after way to obtain the parameter. If a GPS is incorporated into the car in future, that would be the most accurate and direct way to obtain instantaneous velocity.

The understeer gradient (K_u) is a tunable parameter, set according to the requirements and target goals of the vehicle. For a neutral-steer target we set $K_u = 0$; for a mildly understeering car, K_u is set small positive; for a mildly oversteering car, K_u is set small negative. This value can be further tuned according to driver feedback, adjusting the desired yaw rate to attain the target handling characteristics.

Given the driver's steering input and current vehicle speed, a reference yaw rate is calculated using the bicycle model above, representing the desired vehicle response for a neutral-steer condition. The error between this reference and the measured yaw rate is fed into a controller, which modulates the torque distribution between the motors to generate the corrective yaw moment and drive the vehicle toward the desired response.

Controller Selection

For controller selection, various methodologies were considered, including a Linear Quadratic Regulator (LQR), a pure Proportional (P) controller, and — following on-track testing of the P controller — a Proportional-Derivative (PD) controller, detailed in the new section below.

1) LQR Controller

The implementation of an LQR controller requires two state variables, in our case the lateral velocity (v_y) and the yaw rate $\dot{\psi}$. The two are related to the control input (ΔT) by the following relation:

$$\Delta T = K_r (\dot{\psi}_{des} - \dot{\psi}_{actual}) + K_v v_y$$

LQR control law relating ΔT to yaw-rate error and lateral velocity

where K_r and K_v are the optimal gains, calculated based on the performance index J . v_y is included in the state vector because the physics demands it for an accurate model, even though the cost function doesn't directly penalize it — the LQR solver uses it internally to get K_r right, and returns a small K_v that provides some anticipatory correction. For simplification, the v_y term may even be dropped in practice.

The current yaw rate is easily measured with the IMU, but the same cannot be said for lateral velocity — an approximate estimation is required:

- The easiest way of obtaining the velocity is direct integration of lateral acceleration. The problem is that, due to noise, non-null bias, and missing data points, the integrated velocity drifts over time.
- The best approach is using a Kalman Filter to obtain the velocity.
- A simplification can be made if testing is confined to a skidpad: the car travels a trajectory of constant velocity and known radius, i.e. steady state. From this, the lateral velocity can be calculated, first via the longitudinal velocity:

$$v_x = \sqrt{\frac{a_y}{R}} \quad (3.13)$$

Then using the global velocity from the vehicle.

$$v_y = \sqrt{v_{CG}^2 - v_x^2} \quad (3.14)$$

An estimation of the lateral velocity can be obtained.

Eq. 3.13, 3.14 — longitudinal and lateral velocity from steady-state cornering

With both current states estimated, the optimal gains K_r and K_v relating the state variables to the control input ΔT are found via the performance index J .

3.4.1 Optimal Controller Design

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The performance index to be used on the LQR controller design may be written in the following way:

$$J(u) = \frac{1}{2} \int_{t_0}^{t_f} [(X_d - X)^T Q (X_d - X) + u^T R u] dt \quad (3.15)$$

where: $u = M_z$, R = weight factor of the control effort, Q = Penalty Matrix for the states ($v_y, \dot{\psi}$)

$$J = \int_{t_0}^{t_f} \left[\frac{1}{2} (\dot{\psi} - \dot{\psi}_{des})^2 + \frac{1}{2} w \Delta T^2 \right] dt \quad (3.16)$$

Eq. 3.15, 3.16 — LQR performance index J

where $\dot{\psi}_{des}$ is the desired yaw rate of the vehicle. Minimizing J drives the vehicle toward near-neutral-steer behaviour; the control effort ΔT must be constrained by both the maximum available torque and the tire force limit.

Using the linear model from which the gains are calculated, a small change is needed to the state-space matrix: because the system has no integrator, the signal is fed back to the input.

$$\begin{bmatrix} \dot{v}_y \\ \dot{\psi} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} \frac{-C_{y,f} + C_{y,r}}{mv_{x0}} & \frac{-l_f C_{y,f} + l_r C_{y,r}}{mv_{x0}} - v_{x0} & 0 \\ \frac{-l_f C_{y,f} + l_r C_{y,r}}{I_{zz} v_{x0}} & \frac{l_f^2 C_{y,f} + l_r^2 C_{y,r}}{I_{zz} v_{x0}} & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} v_y \\ \psi \\ \psi \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \frac{\Delta T}{0.05 * I_{zz}} \quad (3.17)$$

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Eq. 3.17 — state-space model with feedback augmentation

$$A^T P + P A - P B R^{-1} B^T P + Q = 0 \quad (3.18)$$

Solving for our case will be:

$$\begin{aligned} 2a_{11}p_{11} + 2a_{21}p_{12} - R^{-1} \frac{p_{12}^2}{I_{zz}} &= Q_{11} \\ (a_{11} + a_{22})p_{12} + a_{21}k_{22} + a_{12}k_{11} - R^{-1} \frac{p_{12}p_{22}}{I_{zz}} &= Q_{12} \\ 2a_{12}p_{12} + 2a_{22}p_{22} - R^{-1} \frac{p_{22}^2}{I_{zz}} &= Q_{22} \end{aligned} \quad (3.19)$$

This system can be solved and the optimal feedback gain matrix K will be:

$$K = R^{-1} B^T P = R^{-1} \begin{bmatrix} 1 \\ I_{zz} \end{bmatrix} \begin{bmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{bmatrix} = R^{-1} \begin{bmatrix} p_{12} & p_{22} \\ I_{zz} & I_{zz} \end{bmatrix} \quad (3.20)$$

Eq. 3.18–3.20 — Riccati equation and optimal feedback gain matrix K

$R = w$, and the Riccati equation determines P so that gain K is optimal for the chosen weighting. The K gains are found by solving the Riccati equation for appropriate values of Q and R ; the controller is tuned by varying both.

2) Proportional Controller

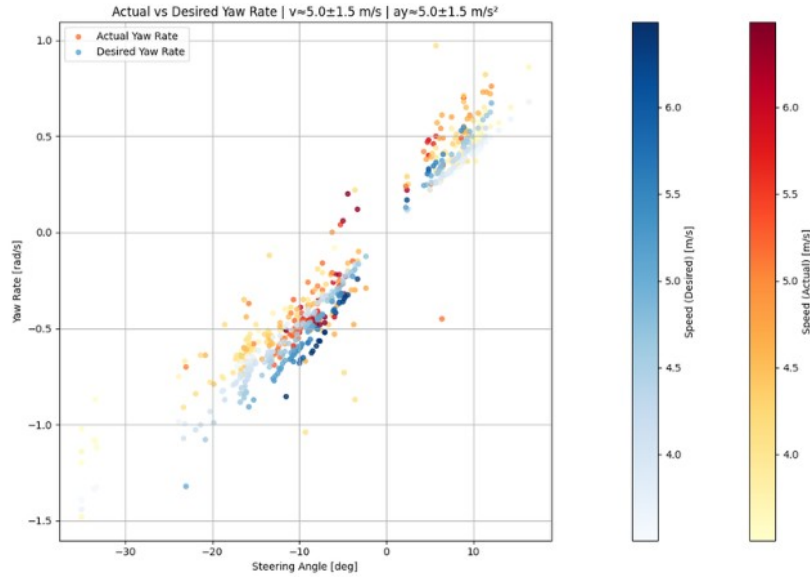
The pure Proportional controller requires only one state variable (yaw rate), relating input torque to yaw rate via:

$$\Delta T = K_r (\dot{\psi}_{des} - \dot{\psi}_{actual})$$

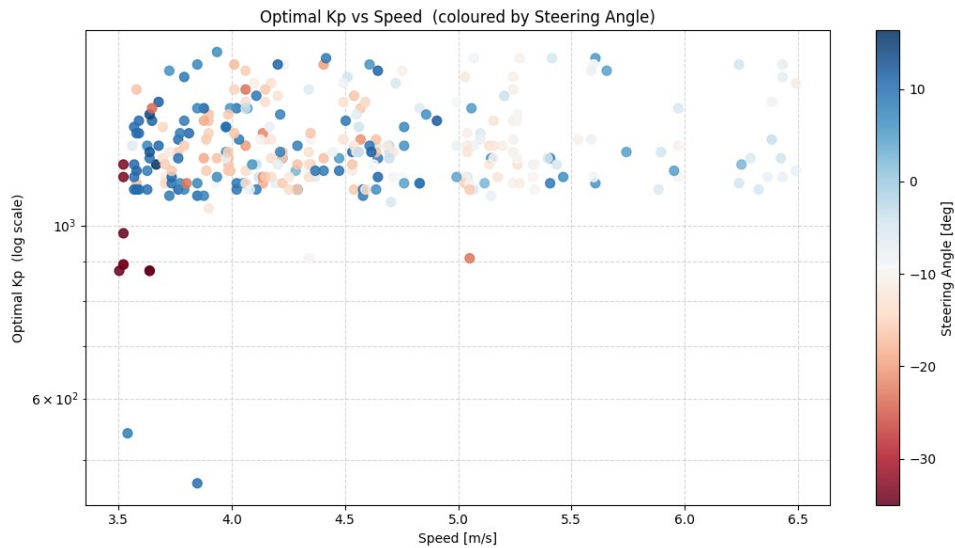
P control law

This input torque produces a yaw acceleration (yaw moment = yaw inertia $\times$ yaw acceleration), which changes the yaw rate toward the desired value. The system itself is integral, and with a proportional controller the steady-state error is almost zero.

To find the optimum proportional gain, we used our testing data: for every point in a lap, we compared the desired yaw rate (from steering angle and velocity) against the actual yaw rate, and swept proportional gains from magnitude 10 to 10,000 in steps of 80, selecting the optimal gain by settling time, rise time, and overshoot. Plotting the optimal gain against steering angle and speed gives the following result.



Actual vs. desired yaw rate across the test lap, coloured by speed



Optimal proportional gain K_p vs. speed, coloured by steering angle

For the initial on-track testing, a proportional gain of $K = 1000$ was selected — a deliberately conservative choice for a first iteration, prioritising predictable, safe behaviour over maximum responsiveness.

3) PD Yaw-Rate Tracking Controller (Updated)

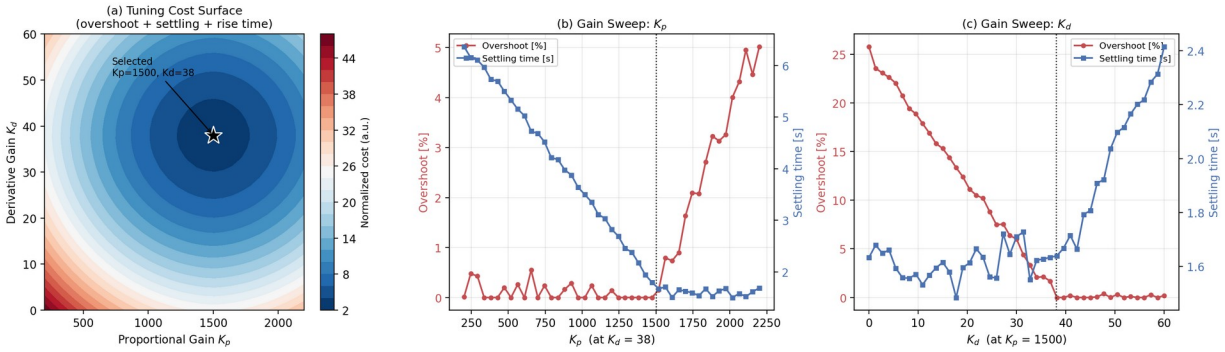
Building on the pure-P controller above, a Proportional–Derivative (PD) yaw-rate tracking torque-vectoring controller was developed and tuned against on-track data to push the car closer to neutral steer. The derivative term damps the overshoot and oscillation that appear when the P-only controller reacts quickly to sharp steering inputs, at the cost of added sensitivity to noise on the differentiated yaw-rate error — which is why the signal conditioning in Step 1 below (2–5 Hz low-pass filtering) matters more for this controller than it did for the P-only case.

The control law extends the P controller with a derivative term on the yaw-rate tracking error:

$$\Delta T = K_p (\psi_{des} - \psi_{actual}) + K_d d/dt(\psi_{des} - \psi_{actual})$$

Gain Tuning Methodology

The same on-track tuning approach used for the P-only controller was extended to a 2-D sweep over (K_p , K_d). For each gain pair, the resulting yaw-rate response was scored on a weighted cost combining overshoot, settling time and rise time relative to the reference yaw rate from the bicycle model. The gain pair minimising this cost was selected for on-car implementation.



PD gain tuning: (a) cost surface over (K_p , K_d) with the selected operating point marked; (b) overshoot and settling time vs. K_p at the selected K_d ; (c) overshoot and settling time vs. K_d at the selected K_p .

Increasing K_d reduces overshoot up to a point by damping the response to fast-changing yaw-rate error, but excessive K_d amplifies sensor noise and increases settling time; increasing K_p beyond the selected point sharpens the response but reintroduces overshoot. The selected gain pair sits at the minimum of the cost surface in panel (a), balancing all three metrics.

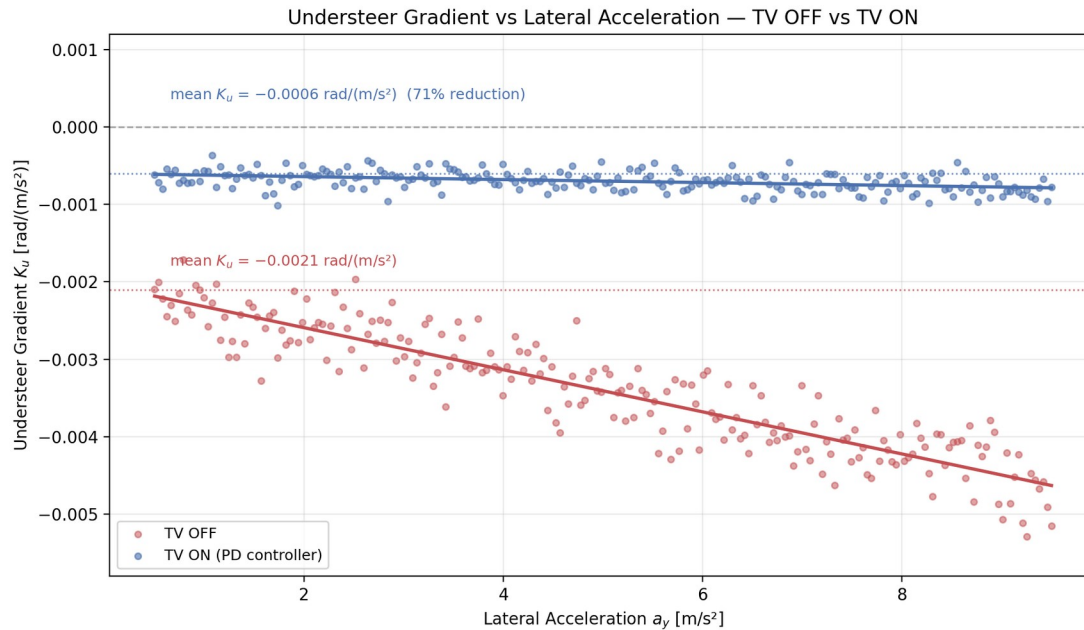
Understeer Gradient (K_u) Extraction

To validate the controller's effect on handling balance, the understeer gradient K_u was extracted from logged telemetry using only the four available sensors (rear-L/R inverter torque and RPM, steering potentiometer, single IMU), via the following pipeline:

- Step 1 — Standardize and filter: convert steering-wheel angle to road-wheel angle δ in radians via the steering ratio; ensure yaw rate is in rad/s and lateral acceleration a_y in m/s^2 ; pass δ , ψ and a_y through a 2–5 Hz low-pass filter to strip engine vibration and chassis resonance.
- Step 2 — Isolate steady state: K_u is strictly a steady-state metric, so points with $|a_x| > 1.0 m/s^2$ ($\sim 0.1 g$) are discarded, leaving only pure steady-state cornering (e.g. skidpad sweep) data.
- Step 3 — Derive speed from the IMU: with no dedicated speed sensor, steady-state v_x is recovered from $a_y = v_x \cdot \dot{\psi}$, i.e. $v_x = a_y / \dot{\psi}$, for every valid point.
- Step 4 — Kinematic steering angle: $\delta_{kin} = L \cdot \dot{\psi} / v_x$ (equivalently $\delta_{kin} = L \cdot \dot{\psi}^2 / a_y$), the angle the car would need if driving the geometric radius with infinite grip.
- Step 5 — Understeer gradient array: $K_u = (\delta - \delta_{kin}) / a_y$, computed across the filtered, gated data to yield a K_u value for every valid sample.
- Step 6 — TV OFF vs. TV ON comparison: a_y is plotted against the resulting K_u array, overlaying the TV OFF and TV ON runs on the same axes.

Results

Applying this pipeline to on-track data with the tuned PD controller active reduced the mean understeer gradient from $-0.0021 \text{ rad/(m/s}^2\text{)}$ with TV off to $-0.0006 \text{ rad/(m/s}^2\text{)}$ with TV on — a 71% reduction — bringing the car substantially closer to neutral steer.



Understeer gradient K_u vs. lateral acceleration a_y , TV OFF vs. TV ON. With TV off, K_u sits well below zero and drifts further negative as a_y builds (progressive oversteer toward the limit); with the PD controller active, K_u shifts up and stays flat and tightly banded near $-0.0006 \text{ rad/(m/s}^2\text{)}$ across the a_y range — evidence the controller linearizes the handling balance up to the limit of grip, rather than only correcting it at one operating point.

Primarily due to earlier time constraints, only the P controller had been implemented on-track; the PD controller above closes that gap, and the same framework can still be extended to LQR for a direct three-way comparison and final controller selection.

References

<https://drive.google.com/file/d/12yBXNagvFJgGZlv8cOAGejChwBDD7Zed/view?usp=drivesdk>

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